INSTRUCTOR GUIDE & ANSWER KEY

SPHERE: MISSION CONTROL

TEACHER GUIDE, LESSON PLAN & SOLUTIONS KEY

Welcome, Educators! This teacher guide is built to help you run a smooth, engaging 60-minute thermodynamics lab where students act as aerospace flight controllers. You'll find curriculum alignment, lab setup tips, theoretical reference tables, and complete answer keys with common student misconceptions.

1. CURRICULUM GOALS & BIG IDEAS

The **SPHERE: Mission Control Digital Twin Lab** blends hands-on physics with live computer simulation. Designed for high school and introductory college physics (NGSS, IB Physics, A-Level), the lab focuses on four core takeaways:

- **How Heat Moves:** Seeing conduction, convection, and radiation in action through real materials.
- **Exponential Cooling:** Understanding Newton's Law of Cooling,

$$T(t) = T_{\text{env}} + (T_0 - T_{\text{env}}) \cdot e^{-kt}$$

  , and what the cooling constant k represents.
- **The "Digital Twin" Concept:** Experiencing how modern engineers run real-time hardware tests alongside predictive computer simulations.
- **Graphing & Data Analysis:** Building dual-curve graphs, calculating absolute errors $|\Delta T|$, and scoring model accuracy.

2. 60-MINUTE LESSON ROADMAP

TIMELINE	LESSON PHASE	WHAT YOU & YOUR STUDENTS ARE DOING
$0–10\ \mathrm{min}$	1. Mission Briefing & Quiz	Hook the class with the extreme temperatures of spacewalks ($+120^\circ\mathrm{C}$ to $-150^\circ\mathrm{C}$). Students complete the quick Pre-Flight Quiz in the Mission Control app to unlock telemetry tools.
$10–20\ \mathrm{min}$	2. Capsule Assembly	Assign crews their insulation setup (A: Bare, B: Cotton, C: Mylar, D: MLI). Show students how to thread the probe through the PG7 gland and check for a snug, leak-free seal.
$20–35\ \mathrm{min}$	3. Live 15-Minute Run	Pour $100\ \mathrm{ml}$ of $80^\circ\mathrm{C}$ water into each jar. Submerge capsules into $0^\circ\mathrm{C}$ ice baths, start the live timer, and log temperatures every $60\ \mathrm{seconds}$ on paper and in the app.
$35–48\ \mathrm{min}$	4. Graphing & Scoring	Students plot their real temperature curve against the digital twin line, sum their $\lvert\Delta T\rvert$ error, and calculate their team's final Correlation Score.
$48–60\ \mathrm{min}$	5. Debrief & Wrap-Up	Lead a quick class reflection comparing all 4 insulation curves. Discuss why Mylar needs spacers, why k changes, and real spacesuit mobility trade-offs.

3. LAB SETUP & SAFETY CHECKLIST

A few minutes of prep before class ensures everything runs smoothly and safely:

- **Hot Water Safety:** Keep water at $80^\circ\mathrm{C}$ using an electric kettle. Have students wear silicone heat grips when pouring hot water.
- **Steady $0^\circ\mathrm{C}$ Ice Baths:** Use a 50/50 mix of crushed ice and water. Make sure plenty of ice floats throughout the $15\ \mathrm{minutes}$ so the bath stays at $0^\circ\mathrm{C}$.
- **Leak Prevention:** Remind students to hand-tighten the PG7 gland dome nut so the internal rubber ring clamps securely around the wire.
- **Probe Placement:** Check that each probe hangs in the center of the liquid and isn't pressed against the plastic jar walls.

4. EXPECTED VALUES & COOLING REFERENCE TABLE

Here is what your student crews should observe after 15 minutes ($T_0 = 80°\mathrm{C}$, $T_{\mathrm{env}} = 0°\mathrm{C}$):

INSULATION TYPE	COOLING RATE (k)	EXPECTED TEMP AT 15 MIN	WHY IT BEHAVES THIS WAY
A. Bare Control	$k \approx 0.150$	$8°\mathrm{C} - 12°\mathrm{C}$	Rapid heat loss through unshielded conduction into water and internal liquid circulation.
B. Cotton Sleeve	$k \approx 0.080$	$\approx 24°\mathrm{C}$	Cotton fibers trap pockets of still air, which is a poor heat conductor.
C. Mylar Foil	$k \approx 0.040$	$\approx 44°\mathrm{C}$	Shiny metallic surface reflects infrared heat radiation back into the core.
D. Full MLI	$k \approx 0.015$	$\approx 64°\mathrm{C}$	The winning combo: cotton blocks conduction, bubble wrap stops convection, and Mylar reflects radiation!

SECTION 5: OFFICIAL SOLUTIONS & TEACHER ANSWER KEY

ANSWER KEY

Question 1: Why doesn't Mylar foil work by itself in cold water?

Sample Answer: Mylar is designed to reflect infrared heat radiation, which is great in the vacuum of space where radiation is the only way heat travels. But in an ice bath, conduction through direct water contact is super fast. Because Mylar is super thin, heat passes right through it by conduction if there is no air spacer (like cotton or bubble wrap) underneath to stop direct contact.
Grading Tip: Look for students mentioning that Mylar reflects radiation, but fails against direct conduction in cold water without a spacer.

Question 2: What does the cooling constant (k) tell us?

Sample Answer: A **smaller** k value means a **better** insulator. In the formula $T(t) = T_{\text{env}} + (T_0 - T_{\text{env}}) \cdot e^{-kt}$, when k is small, the exponent $-kt$ stays close to 0, which keeps e^{-kt} close to 1. This means the temperature stays higher for longer and cools down much more slowly.
Grading Tip: Students must state that smaller k = better insulation and explain that it slows down cooling.

Question 3: How does Multi-Layer Insulation (MLI) work together?

Sample Answer: Each material tackles a different type of heat loss:

- **Cotton** traps still air to stop *conduction*.
- **Bubble wrap** seals off air pockets to stop *convection* currents.
- **Mylar foil** reflects infrared *radiation* back inside.

Together, they block all three heat escape routes at once, keeping the water above $60°\text{C}$!
Grading Tip: Full credit requires mentioning how the materials address conduction, convection, and radiation.

Question 4: Where could experimental errors come from?

Sample Answer (Accept any two sensible answers):

1. The sensor tip was touching the cold plastic jar wall instead of hanging in the middle of the water.
2. Water leaked into the jar through a loose cable gland.
3. The ice bath warmed up above $0°\mathrm{C}$ during the $15\ \mathrm{minutes}$.
4. The water inside the jar wasn't stirred, so warm water rose to the top and cold water sank to the bottom.

Question 5: What happens if your ice bath warms up?

Sample Answer: Students should measure the real bath temperature (e.g. $4.5°\mathrm{C}$) and type that new number into the T_{env} box in the Mission Control simulator. Updating the environment temperature makes the theoretical curve match the real classroom conditions.
Grading Tip: Award credit if they mention updating T_{env} or the environment temperature input in the software.

Question 6: What if we double the water volume?

Sample Answer: Doubling the water from $100\ \mathrm{ml}$ to $200\ \mathrm{ml}$ doubles the mass of hot liquid ($Q = m \cdot c \cdot \Delta T$), meaning it holds twice as much thermal energy. Because there is more heat to lose through roughly the same surface area, it cools down **slower**, and the cooling constant k **decreases**.
Grading Tip: Look for: cools slower and k decreases.

Question 7: Spacesuit Design Trade-Offs

Sample Answer: You can't just add 50 layers because the spacesuit would become too heavy, bulky, and stiff. Astronauts wouldn't be able to bend their elbows or fingers during a 7-hour spacewalk. Also, human bodies generate lots of body heat when working hard—too much insulation would trap that heat and cause dangerous overheating without heavy water-cooling systems.
Grading Tip: Accept answers discussing joint flexibility/mobility, suit weight/bulk, or body heat buildup.

Question 8: How can you improve your Correlation Score?

Sample Answer:

1. Tighten the cable gland securely to prevent any cold water leaks.
2. Make sure the probe tip is suspended exactly in the center of the jar.
3. Swirl the jar gently before reading to keep the water mixed evenly.
4. Take readings at the exact second mark on the timer.

6. GRADING RUBRIC & CHECKLIST

CRITERIA	EXCELLENT (90–100%)	GOOD (75–89%)	NEEDS WORK (<75%)
Data Logging	All 15 minutes filled with accurate readings, absolute differences, and helpful notes.	Completed with only 1–2 missed minutes; minor calculation mistakes.	Multiple blank rows; missing calculations.
Graphing	Clean, accurate points plotted with a smooth line; dashed theoretical curve clearly shown.	Points plotted with minor curve scaling inaccuracies.	Points not connected; axes mislabeled or missing lines.
Analysis Questions	Clear, thoughtful answers explaining conduction, convection, radiation, and MLI synergy (Q1–Q8).	Good understanding of heat transfer with minor details missing.	Confuses conduction with radiation; incomplete answers.